Tracking Events Through a News Stream: A Filter That Works, an Attribution Layer That Does Not

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Revision of 2026-08-19. Companion to `PAPER_0_FOUNDATION.md` (the extraction substrate) and `JOINT_IE_SCALING.md` (the within-document instantiation of the same thesis). The build record — codebase attachment points, CLI, training plan, decision log, generator specification, test protocol, and the negative-sampling implementation notes — is kept verbatim in `EKF_MHT_BUILD_RECORD.md`.

Abstract

A real-world event reported across a stream of news documents has a state that evolves: a death toll rises, gets revised downward, and is reported by sources that disagree, censor ("at least 12"), and lag. We build the full pipeline — synthetic stream generator, LLM realizer, schema-driven extraction, normalization, and a per-role Extended Kalman Filter — and validate it end to end on held-out synthetic data, where fine-tuning the extractor closes **75% of the gap to the structured-observation ceiling**.

Then we run it on a real event, and it loses to repeating the last reading.

This paper reports that outcome and what it localizes. The programme's question has two halves — **track** an evolving quantity, and **diarize** observations into the right event stream — and only the first is built. What ships for the second is hard assignment on a string key, not the specified multi-hypothesis association. Every real-event failure we have measured is in that second half: a 1999 earthquake's death toll tracked as a 2023 figure, one of two affected countries never recovered at all, and a hurricane's toll contaminated by three unrelated storms quoted in the same articles. A magnitude-based scope gate recovers a **9× improvement** on the event it was designed against but **splits held-out** — 3.7× better on the contaminated stream, 2.3× worse on the clean one — and the reason is diagnostic: without a declared scope hierarchy, "a larger scope" and "the largest part" are indistinguishable from the numbers alone. We then declared one, and the reference it enables is itself a measured negative.

Before building the multi-hypothesis association the design specifies, we price it, and the first price is wrong. A two-way oracle — assign each observation to its own place or to the national total — improves the shipped gate by only 0.055, which reads as "association is not what is missing". But that oracle cannot *reject*, so a figure belonging to no scope in this event has no correct home; give it one and the ceiling moves to **0.111 (18.8%)**, double the figure, with about half of it needing a null hypothesis and half not. We then build the cheapest piece that delivers one — track birth by innovation gating — and **it loses to the fixed magnitude ratio it was meant to replace**, degrading the national stream 6.7× while it does so, because judging a stream against its own track is circular in exactly the way §5's scope reference was.

The residual underneath is 4.7% cross-event contamination against which no decode-side signal has worked, and the corpora offer no purchase on it: **0.0% of training documents contain a figure the model is supposed to leave alone.** So we build that corpus, withholding an interfering event's records while keeping its text, and train it. The suppression is real: it removes 15 of 20 large false positives and more than halves the ungated error. **Then a declared per-event plausibility ceiling — one threshold, no model, no training — beats it**, because the large false positives were never other storms' tolls. They were Asheville's population, FEMA flood-insurance policies, power crews, and years read as death tolls. Both genuine cross-event figures survive muting *and* the ceiling.

The remaining candidate — embedding each event's own trigger-and-argument span and matching against live filters — could not be run at all, because no extractor we have emits that input on wire copy. Counting only corpora that bind an argument to a trigger, the boundary base has **798 English rows against 20,884 Chinese**, so its argument score is very nearly a Chinese-only number. **The critical path is therefore the extractor, not the tracker** — a conclusion three built-and-failed association mechanisms were needed to reach.

We also report that our own strongest prior result does not reproduce. An ablation concluding the filter's advantage *widens* under noise and censoring was measured on synthetic streams generated by the dynamics the filter models. On real revision data the advantage is real, reliable, and **1–2%**. And the cached observation set behind every Helene number here — including the 9× scope-gate result — turns out to be unreproducible from any committed state of the repository, which we document rather than quietly re-baseline.

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1. The question, stated as it was actually asked

Can a filter track real-world events reported in text, and separate them into the right streams?

Two halves, co-equal, and the second is not downstream of the first:

half	mechanism	status
track — recover an evolving quantity from noisy, censored, lagged reports	Extended Kalman Filter (§2)	built; validated on synthetic streams
diarize — decide *which* event stream each observation belongs to	multi-hypothesis association	**not built**

"Diarize" is borrowed deliberately from speaker diarization. It is not "who spoke when" but *which event is this figure about, over time*, and naming it that way makes obvious that the two halves fail independently and that the second can silently destroy the first.

The association half is unbuilt, and this should be stated plainly rather than discovered by a reader. The design specifies association as gate → Hungarian assignment → top-K hypotheses → track birth and death. What ships is **hard assignment on an observable string key**, feeding one single-stream filter per key: no hypothesis enumeration, no deferred decision, no track birth or death. The only Hungarian solver in the repository is inside the extraction model's record loss, matching predictions to gold during training — unrelated to tracking.

That reframes every result below. The failures reported in §4 and §5 are failures of a **placeholder**, not of multi-hypothesis tracking. The mechanism the design names for exactly this problem has never been tested, so the programme's central question is not yet answered — it is, so far, unasked.

2. Method

Granularity. Situation level, not event-instance level. A static instance has weak dynamics and the filter degenerates to fusion.

State. Per track: a pooled embedding, the continuous quantitative arguments (normalized counts, amounts, dates, magnitudes), and a salience term, with covariance.

Dynamics. Slow-drift embedding; monotone, random-walk or parametric process for the quantities; decay on salience, which provides track death. As process noise goes to zero the filter degenerates to optimal static fusion, so the single-document case is safe by construction — a mild win over naive merging, never a loss. Dynamics are inert by default and engaged only by evidence of change.

Measurement. Each document's extraction is a noisy and often censored observation — "at least 12", "roughly $3M" — which is why the filter is *extended*: the measurement function is linearized.

Gating. A one-sided innovation gate, plus a learned mixture-of-experts router blending the local per-document read against the tracked estimate. The router is what withholds the filter in exactly the regime where it would degrade a static read.

The pipeline under test. Synthetic stream generator → LLM realizer producing news-like prose → schema-driven extraction → confidence cut → normalization → per-role filter. Each stage is separately ablatable, which is what made the diagnosis in §3 possible.

3. The synthetic arc: extraction is solvable

Working backwards from a structured-observation ceiling of 0.115 normalized RMSE — the score achievable if extraction were perfect — the zero-shot extractor reached 0.291. A probe localized the gap to extractor precision and confidence calibration rather than to normalization, and predicted that fine-tuning would close it. Fine-tuning on 29,198 casualty-structure examples:

held-out test	zero-shot	fine-tuned
role precision	0.627	**0.914**
role recall	0.906	**0.965**
value exact (number binding)	0.991	**1.000**

end-to-end (normalized RMSE; ceiling 0.115)	zero-shot	fine-tuned
best (confidence cut 0.99)	0.291	**0.165**
no confidence cut	1.16	**0.193**
the weak `missing` role	0.458	**0.122**

The fine-tune closed **~75% of the gap to the ceiling**, exactly as predicted, and removed the dependence on a confidence cut — uncut performance went from unusable to 0.193. Number binding is perfect.

The caveat that turned out to matter most: this was fine-tuned and tested on the same synthetic distribution. Real news was left as the generalization test, and it is where the arc breaks.

4. The first real event, and the first real defeat

Türkiye–Syria 2023, pre-registered before running. Ground truth is 16 daily points scraped from a single news tracker page via an archive service; search summaries were checked and found to **contradict each other on dates**, so they were rejected as a source.

On real news the trivial baseline wins: `est_last_value` 0.208 against the filter's 0.136 — reported here in the direction the metric runs, where the filter's 0.136 is the better score, but the finding is that the ranking reverses relative to every synthetic benchmark, and the margins are small enough that neither is a good result.

Three failures, none of them in the component the previous three sections optimized:

- **A 1999 death toll is tracked as a 2023 figure.** The 17,500 dead of the Izmit earthquake, quoted in an article's historical background section, enters the stream as a current observation — in every configuration.
- **Syria is never recovered.** The association key is computed per *document*, outside the envelope loop, so a two-country event collapses to one country.
- **Extraction is not the problem.** It reads the real trajectory nearly point for point.

So the bottleneck is attribution, not filtering and not extraction. That is the finding, and it redirected the programme away from further extractor work.

A method rule learned the hard way. A pre-registered prediction was scored a *success* by range-normalized RMSE while 12 of 20 readings came from a 1999 earthquake: the metric is blind to contamination that lands mid-range. Always report the trivial baseline beside the filter — it would have caught this immediately.

5. The scope gate: a 9× win that splits when held out

Reporting mixes scopes: a national total and its state components appear in the same sentence. Filing the national figure under whichever state is nearest — what shipped — is plainly wrong.

A magnitude gate gives each observation a scope test against a reference. On Hurricane Helene, the event it was designed against:

ratio	national stream	per-state mean
off	0.402	5.247
2.0	0.316	**0.591**

Per-state error falls **5.247 → 0.591**, a 9× improvement, while the national stream *improves* rather than paying for it, and the effect is flat across ratios 1.5–2.5 rather than being a knife-edge setting. **Control:** removing the same number of observations at random over 40 trials gives 4.427, so the gate is *selecting* rather than thinning the stream into looking better.

Held out on Türkiye–Syria at the ratio fixed from Helene, it splits:

	Türkiye	Syria	mean
off	**0.228**	3.401	1.815
gate @2.0	0.522	**0.923**	0.723

Syria was 65% contaminated — 11 of 17 values are Türkiye's tolls — and improves 3.7×. But **Türkiye, already clean, degrades 2.3×.** The reason is structural rather than a tuning failure: Türkiye–Syria declares no scope hierarchy, so the reference falls back to the running maximum across streams, which Türkiye's own values dominate. Türkiye is judged against a reference it defines itself. The gate rerouted a value that was Türkiye's *true* reading at that time.

The diagnostic finding: the gate needs a declared scope hierarchy, not a magnitude. Without one, "a larger scope" and "the largest part" are genuinely indistinguishable from the numbers alone — Türkiye's 41,000 filed under Syria and Türkiye's 41,000 filed under Türkiye look identical to any ratio.

And the obvious fix for it was built, measured, and does not work. Both events now declare containment explicitly — `rollup.json` carries a `hierarchy` block naming the aggregate and its parts — which enables the reference the diagnosis calls for: judge a part against its **implied maximum**, `aggregate − sum(other parts)`, rather than against a bare magnitude. That is the right shape for the dominant-part problem on paper. Turkey against an implied max of 46,800 − 5,800 = 41,000 sits exactly at its ceiling and is kept, while the same 41,000 filed under Syria faces an implied max of 5,800 and is rerouted.

On Helene it is **much worse than the plain aggregate reference: 2.590 against 0.591.** The parts are themselves contaminated, so their raw sum exceeds the whole and every implied maximum clamps to zero; a two-pass version recovers most of that but Florida still sits at 9.437. And Türkiye–Syria — the event that motivated the refinement — **still cannot test it**, because an implied maximum needs *independent* observations of the whole and that feed has none. Declaring the hierarchy does not manufacture the data.

So the diagnosis stands and its first implementation is a measured negative. Plain aggregate remains the recommended reference wherever an aggregate stream exists, and where none exists no gate is possible at all. Detail in `EKF_MHT_BUILD_RECORD.md` §25.5.

Honest scorecard. A 9× win with a clean control on the event it was designed against; held out, 3.7× better on one stream and 2.3× worse on the other. Both belong in any writeup, and the Helene number alone is the misleading half. The ratio was also chosen after seeing Helene's contaminated values; the plateau across 1.5–2.5 mitigates that without removing it. And 0.591 is nine times better than catastrophic, not good in absolute terms.

What the gate does not address, visible in the same audit: North Carolina's 1,400 is *Hurricane Katrina's* toll quoted inside a Helene article, its 250 is a typhoon, and its 230 is Hurricane Milton. That is **cross-event** contamination, not scope. A magnitude gate removes these for the wrong reason — because they are large, not because they belong to another event — so it would keep any cross-event figure that happened to be small.

6. Results that did not reproduce, and results that were negative

Reported at the same weight as the positives, because collectively they are what localizes the remaining work.

6.1 Our strongest prior claim does not reproduce

An earlier ablation concluded the filter's edge *widens* under unreliability and censoring, and that conclusion was standing as the answer to "does the filter earn its keep". On the only real revision data available — a hurricane's North Carolina toll falling 123 → 102 → 96 as deaths were reclassified — it does not hold:

regime	filter	last_value	gain	filter wins
noise 0.05, report 80%	0.094	0.095	+1.3%	191/240
noise 0.10, report 60%	0.189	0.192	+1.5%	205/240
noise 0.20, report 40%	0.322	0.327	**+1.8%**	217/240
noise 0.30, report 30%	0.499	0.504	+1.0%	222/240
noise 0.40, report 20%	0.628	0.634	+0.8%	217/240

A revision is the case where `last_value` looks best — it adopts the new figure at once while a filter lags — so the filter can only win by smoothing noise the baseline chases. **It wins reliably**, in 80–93% of trials, on the first benchmark in this project whose baseline is not an oracle. **And it wins by 1–2%**, roughly flat, *shrinking* at the hardest setting rather than widening.

Why the earlier result was wrong: it measured synthetic streams generated by the same rise/decay dynamics the filter models, so the filter was matched to the process by construction. Real trajectories — plateaus, reclassifications, a genuine downward step — are not that process. *A dynamics model validated on data generated from it is not validated.* The case for the filter must rest on something other than the harder-regime argument, most plausibly multi-source disagreement, which no benchmark here has tested because every feed so far has a single source.

6.2 Aggregates cannot constrain their parts

Making the state a vector so a national total becomes a linear observation over states is the theoretically right move — a Kalman filter fuses "120 in North Carolina" and "227 across six states" natively, where any clustering method would work in value space and get it backwards. Tested against a parts-only filter on real per-state trajectories, it loses at every realistic reporting density and only reaches parity at 80%:

per-state report rate	parts-only	vector	vector wins
10%	**0.4348**	0.6085	4/40
35%	**0.2292**	0.2860	10/40
80%	0.1556	**0.1520**	30/40

Right in principle, and it does not pay off at the densities real feeds provide.

6.3 Restructuring the text does not fix attribution

The proposal was a summarizer emitting self-contained bullets so number-to-place binding becomes local. Tested with hand-written bullets first, so a negative would cost nothing:

arm	correct	false positives	fabrications
raw text	**3/5**	1	0
free bullets	2/5	3	**2**
extractive bullets	**3/5**	1	0

Restructuring is neutral at best and the free variant fabricates. **The motivating example does not occur in the corpus at all** — the real failures are non-casualty numbers and cross-event leakage, not aggregate splitting. There is also a structural tension: the summarizer's highest-value act is normalizing prose into digits ("they died together" → "2 people died"), which is precisely what a verbatim-number guard must reject.

6.4 Guide-filtered negative sampling: wired, tested, negative

The residual confusion after description tuning is a genuine hard negative — "N people killed" against "N people evacuated", both counts of people, separated only by the verb, accounting for 11.2% of true death tolls. This is the case a guide-filtered loss exists for: the competing type is genuinely correct sometimes and genuinely wrong here, so a loss that penalizes it unconditionally would teach the wrong thing.

Implemented as a query-axis veto with a guide model and run as a two-arm A/B on 84,280 records, both arms swept to their own threshold:

metric	control	guide veto	delta
entity strict	0.5858	0.5610	**−0.0248**
relation strict	0.1439	0.1108	**−0.0331**
event_argument fair	0.5786	0.5702	−0.0084
event strict	0.3433	0.3443	+0.0010

Negative on seven of eight metrics, including the argument axis the veto exists to sharpen. The control validates the setup by reproducing the historical reference from scratch. Against a measured run-to-run floor, the entity loss survives and the relation loss does not, so the honest statement is that the veto is neutral-to-harmful and is not the mechanism for this failure.

Wiring it did surface two ways it would have been **silently inert**, which is the transferable part. A sample's query axis holds only the types its own record declares, and just 0.23% of records name a competing count type natively — so the cross-record rivals the method exists for are never present unless something injects them, and the veto would have run, cost nothing and changed nothing. Separately, the veto resolved each candidate's reference by position in the candidate list, which is only valid for a shared candidate pool and is false for the per-query default that ships.

7. Where this leaves the programme

Built and validated: the filter, its measurement model, the learned gate, and the text-to-observation pipeline, end to end on held-out synthetic data at roughly 1.4× the structured-observation ceiling.

Built and honestly weak: the filter's advantage over repeating the last reading is 1–2% on real revision data, and the argument that it widens under hard regimes does not reproduce.

Not built: multi-hypothesis association — the half every real-event failure lives in. The scope gate of §5 is the first concrete instance of that layer doing useful work, in its simplest possible form and without a new model.

7.1 How much is left for better association — measured, not argued

The natural conclusion from §4 and §5 is "build the association layer properly," and MHT is the specified answer. Before spending on a hypothesis tree, a cost matrix, Hungarian assignment and track management, we measured what *any* better association could be worth. The measurement needs no new model: assign every observation to the scope it actually fits, using ground truth. It is a ceiling, not a method.

	per-place mean nRMSE
no gate	5.247
shipped scope gate	0.591
oracle hard association (uses ground truth)	**0.537**
headroom for any better association	**+0.055** (9.3%)

Perfect association buys 0.055 — and that number prices the wrong thing, which we found only by trying to spend against it. The oracle is *two-way*: every observation goes to its own place or to Total, so a figure belonging to no Helene scope has no correct home and it scores Katrina's 1,400 exactly as badly as the shipped gate does. The tell was already in the table below, read as a curiosity rather than as a defect in the instrument.

MHT's track birth/death **is** a null hypothesis, so give the oracle a reject option and sweep the tolerance rather than fixing it:

tol	kept	per-place mean
2.00	100	0.533
1.00	100	0.533
0.50	**85**	**0.480**
0.25	76	0.499

The corrected headroom is 0.591 → 0.480 = +0.111, 18.8% — double the figure MHT was rejected on. It splits almost evenly: 0.591 → 0.537 is reassignment (+0.055, mostly Tennessee) and 0.537 → 0.480 is the reject option (+0.057), so about half the prize needs a null hypothesis and half does not. Still a ground-truth ceiling, still one event, and the tolerance is tuned and non-monotone, so there is no plateau to hide behind.

The per-place breakdown sharpens it. The gate already **beats** a perfect two-way assignment on Florida (0.704 vs 0.734) and South Carolina (0.365 vs 0.558), because it has a third option the oracle lacks — *drop*. Florida's 300 is not a casualty figure and North Carolina's 1,400 is Hurricane Katrina's toll: neither belongs to *any* Helene scope, so no assignment scheme can place them correctly. Only Tennessee is a genuine association gap (0.817 vs 0.320), and its contaminants — 32, 32, 32, 36, 50 against a truth of 18 — are instructive: too large for the state, too small to look national, so no magnitude test can catch them. That is the real case for richer association.

7.2 The cheapest piece of MHT, built and lost

The reject option is where the prize is, and it is also the cheapest piece: track birth needs no cost matrix and no hypothesis tree. So we built it. Tracks advance jointly in time order, each observation is tested by normalized innovation against its candidate tracks, and one that gates out of all of them is born into its own and leaves these streams. No ground truth.

Nothing swept beats the fixed magnitude ratio it was meant to replace:

	Total	per-place
no gate	0.402	5.247
symmetric birth, own + aggregate (`q_rel` 0.20, the filter's own)	0.555	1.059
symmetric birth, tuned (`q_rel` 2.00)	2.115	0.636
one-sided birth, tuned	2.115	0.608
aggregate-only reference (`q_rel` 0.20)	0.387	0.624
shipped magnitude scope gate	**0.316**	**0.591**
three-way oracle (ground truth)	0.308	0.480

Both columns matter, and the second one alone would have flattered this. The two tuned arms buy their per-place improvement by dumping junk into the aggregate — Total 2.115 against the gate's 0.316, a 6.7× degradation of the national stream, which is the one measurement §4 calls honest. That is exactly the failure the shipped gate's three-outcome design was invented to prevent: an earlier two-way version rerouted every reject to `__aggregate__` and destroyed that stream. This associator reproduced a bug the project had already fixed.

Two causes, both measured. **Judging a stream against its own track is circular** — every contaminant the track accepts moves the reference the next test uses — and removing the self-reference is worth more than every other knob combined, 1.059 → 0.624. That is the same failure the implied-maximum reference hit on Türkiye in §5, so it is now two independent mechanisms defeated by one cause. And **the innovation is not informative about scope on a rising toll**: at the filter's own dynamics the tracks are too tight to admit real growth, Georgia keeping only 2 and 3 against a truth peak of 34, and the sweep must loosen them tenfold before real rises survive — by which point only one to four observations are ever born. Birth is never the lever; the rerouting is.

This does not kill deferred assignment; it is the first evidence *for* it. Hard assignment commits early and poisons its own reference, which is precisely what keeping rival hypotheses alive exists to prevent. Before this run that was a design preference.

And it sets up the section that follows. The observations the reject option would remove are substantially the same observations the next section counts as cross-event contamination — Katrina's 1,400 is both. There are therefore two routes to them: a decode-side one (M4, now measurement-implicated but a large build against a 0.111 ceiling on a single-source feed) and a data-side one. The data-side route is the subject of §7.3, and the reason it goes first is that it is already built and instrumented while M4 is not.

7.3 What the residual actually is

A context audit of all 106 observations puts the remainder somewhere else entirely: 82.1% genuine Helene casualties, **4.7% belonging to another event** (Katrina's 1,400, a typhoon's 250, Milton's 230, Bosnia's 16, Hurricane John's 2 in Mexico), 3.8% non-casualty numbers (speeds, rainfall), and a 9.4% unclear tail.

Cross-event contamination carries the *large* values, so it does the most damage per instance — and the scope gate removes the large ones for the wrong reason, because they are big rather than because they belong to another storm. It therefore keeps any *small* cross-event figure, which is exactly what happens with Bosnia's 16 and Mexico's 2. Bosnia is structurally invisible to every signal tried so far: it is a *place*, not a named storm, so nothing keyed on storm names can see it.

This audit describes the **archived** observation set, and that set can no longer be regenerated (§9). The percentages below should be read as a description of one frozen artifact rather than as a stable property of the pipeline.

So the indicated next step was neither MHT nor more extractor fine-tuning, but negative supervision on event identity. The evidence for a training-data gap rather than a decode gap is that binding collapses from 1.000 to 0.369 the moment documents become multi-event, and no decoder change has moved it. Every training corpus is complicit — measured over 20,000 records per corpus, **0.0% of training documents have zero records**, so the model has never once been shown a casualty figure it is supposed to leave alone.

We built it. §7.4 reports what happened.

7.4 Negative supervision: built, and beaten by a one-line threshold

`casualty_loc_muted` keeps an interference snippet's *text* and withholds its *record*, so its figures become negatives for the same queries — the first casualty corpus in which the model is shown a figure it must leave alone. Two arms were trained identically for four epochs, differing only in the corpus, with the control's own binding numbers measured beforehand rather than inherited from a superseded checkpoint.

The treatment took. On the blind test, precision rose and recall fell (0.8119/0.8182 → 0.8273/0.7754) — suppression, pre-registered as the expected shape because the test split carries gold on every interference snippet. On Helene it removed **15 of the control's 20** large false positives and cut the ungated per-place error from **46.844 to 19.822**.

Then the cheapest possible alternative beat it. Auditing what those large values actually were found that none of them is a Helene death toll:

class	examples
resident population	94,000 (Asheville) · 19,000 (Boone) · 3,100 (Saluda)
people, but not casualties	1,500 active-duty troops · 8,000 power crews
not people at all	**129,933 FEMA flood-insurance policies** · 15,000 wellness checks · 1,100 churches
year read as a toll	1,916 ("since 1916") · 2,004 ("**In 2004**… four people were killed")
cross-event	Katrina's 1,400 · Maria's 3,000

The 15,000 is the sharpest: its sentence exists to warn against this exact error — "*that was mistakenly interpreted as meaning 15,000 people were missing*" — and the extractor made it anyway. The 94,000 is emitted as `dead` *and* `injured` *and* `missing` at confidence 1.0.

A **declared per-event plausibility ceiling** — Helene killed on the order of 230, so a five-figure figure in any of its streams is not a casualty count — needs no model, no training and no GPU. Swept rather than chosen:

ceiling	production model	control	muted
off	378.809	46.844	19.822
20,000	**18.287**	26.961	19.822
2,000	18.190	**5.853**	6.194
1,000	18.190	5.057	6.194

Dropping the single 94,000 is worth **20×** on the production model. And at a ceiling of 2,000 — about nine times the true toll, so generous rather than tuned — **the control beats the muted arm on both ungated (5.853 vs 6.194) and gated (3.336 vs 3.729)**, while carrying 81 *more* observations. The ceiling removes only junk; muting removed genuine signal alongside it.

The arm is therefore superseded, not vindicated. Its pre-registered guard was the ungated comparison, and it passes only against an *undefended* control — a configuration nobody would ship. What stands is that the suppression is real and learned. What does not stand is that it is worth having.

And it fixed the class it was not built for. Both cross-event tolls survive muting, and survive the ceiling too: at 2,000 Katrina's 1,400 is a perfectly plausible magnitude, and a ceiling set low enough to catch it is no longer a plausibility test but the magnitude gate again, rejecting a figure for being large rather than for belonging to another storm. The same holds for the 1,500 troops and 8,000 crews — living people in the affected area, wrong for a reason no ceiling can see and no entity-type check can reach. **Cross-event contamination remains unsolved.**

That is the paper's main practical contribution, and reaching it took a real event, a wrong price, and two builds that lost: the obvious next build was first rejected on a ceiling measuring the wrong thing, rebuilt in its cheapest form and lost on its own terms (§7.2), and the data-side alternative was then beaten by a threshold.

7.5 The critical path moved to the extractor, and it is being rebuilt

Added 2026-08-20, after §7.4. With negative supervision superseded and MHT re-priced, the remaining candidate is a **span-embedding router**: take `min(start)..max(end)` over an event's own trigger and arguments, embed that block, and match a new observation against the live filters. It is the first proposal that *builds* a representation rather than assuming one, and the block is per-event and local — which is the discourse-attachment job that proximity and type both failed at. On the Katrina passage the only named event the extractor finds is `Hurricane Katrina`, so the block is Katrina-local, not Helene-local.

We could not run it, because nothing available emits the input.

The span architecture emits a *bag* of triggers and labels every one with the same role. No threshold works: below 0.4 the Katrina block is the bare name, missing the 1,400 it should bind; the Helene block runs the length of the passage and swallows Katrina; at 0.5+ Katrina yields nothing. The boundary/mmBERT base yields nothing above threshold 0.3 on English disaster copy and nonsense at 0.1 — trigger `"remote"`, with both `dead` and `location` bound to `"Helene decimated"`.

The cause is arithmetic, and finding it required correcting a claim we had just made. A first pass read the model card's row counts and said 68% of its event supervision was Chinese. That was wrong: DocEE, ChFinAnn and DocFEE are not stored as events at all — they are `entities` + `classifications`, so the doc-level type is a class label and the arguments are NER spans, and counting them as event training flattered both sides of the ratio. Counting only corpora that bind arguments to a trigger:

	English	Chinese	English share
the base as built	**798** (CASIE alone)	20,884	**3.7%**
every corpus available	39,783	20,884	65.6%

MAVEN and Mendeley are trigger-only. So an argument F1 of 0.506 is very nearly a Chinese-only number, and English trigger→argument rests on 798 examples — which no threshold reaches.

A cold-start rebuild is training now: 189,284 records, a 50× increase in English trigger→argument, the Chinese corpora kept because they are why the argument head works at all. Its gates are on AP prose rather than held-out DocEE, for the reason this section exists. The declared risk is that 72% of the new English data is synthetic, on a programme whose recurring failure is in-domain-good and real-news-zero.

So the critical path is no longer the tracker or the association layer. It is the extractor, and that is a conclusion three failed association mechanisms had to be built before anyone could reach.

8. Limitations

- **One real event, one source per feed.** Both real evaluations draw ground truth from a single tracker page. The multi-source disagreement case — the most plausible remaining argument for a filter over a last-value baseline — is untested.
- **The scope-gate ratio was chosen after seeing contaminated values** on the event it was tuned against. The plateau across 1.5–2.5 mitigates this; it does not remove it.
- **Synthetic validation was circular in a specific way** (§6.1) and any future synthetic benchmark must generate trajectories from a process the filter does *not* model.
- **Cross-event contamination is unaddressed.** It is a distinct failure from scope, needs instance-level rather than type-level discrimination, and the one mechanism tried for it (§6.4) was negative.
- **A genuinely blind real test remains outstanding.** Both events evaluated here predate the model's knowledge cutoff.
- **The Helene observation set behind every number here is a cached artifact that cannot be regenerated** (§9.1). Comparisons among the published figures remain valid; placing a new model on the same scale does not.
- **Every association result here is bounded by an extractor that cannot express the alternative.** §7.5: no available model emits per-event trigger-and-argument spans on wire copy, so the span-embedding router is untested rather than refuted. A rebuild is in flight.
- **The plausibility ceiling of §7.4 is prior knowledge, declared per event.** It is not learned, it does not transfer to an event whose scale is unknown in advance, and on a feed with no large false

positives it does almost nothing (5.247 → 4.283 on the archived set).

9. Reproducibility

9.1 The Helene observation set cannot be regenerated

Every Helene number in this paper — 5.247 → 0.591, the oracle ceilings 0.537 and 0.480, and the M5 sweep of §7.2 — is computed from one cached artifact, `datasets/helene2024/_cache/tracked_rollup.json`, written 2026-08-10. That artifact cannot be reproduced from any committed state of the repository.

Established by elimination, 2026-08-20. The extraction model is unchanged on the Hub since before the run; the CLI defaults are identical; a worktree at the contemporaneous commit differs from current code by three observations out of ninety. The archived grid identifies one undocumented flag (`--grid-step 12`). The blocker is elsewhere: **the `--rollup` flag did not exist in any commit before the artifact was written**, and the rollup file itself was not in the tree either — both were uncommitted working-tree state, committed later in a form that does not reproduce it. There are no commits in the intervening window.

The signature is unambiguous. The archived run assigns every observation to a scope; every reproduction leaves half of them keyed `unknown`, because the record head returns no location. Reproductions also put a **94,000** into Tennessee, where the archived run's largest value anywhere is 3,000.

What this does and does not cost:

- **Comparisons among the published figures stand.** They all read the same frozen artifact, so they are mutually consistent.
- **No newly trained model can be placed on that scale.** This blocked the pre-registered guard of §7.4 and forced a fresh baseline under one recorded invocation.
- The remedy is a new reference, not archaeology. The figures above should be read as belonging to a superseded observation set.

`run_pipeline.py` now writes its complete argument vector and git commit — with a `-dirty` marker — into every output. That marker is the load-bearing part: the archived run came from a tree matching no commit, and nothing in the artifact said so. This is the second time provenance has blocked this programme; the Türkiye–Syria re-extraction is stalled the same way (`EKF_MHT_BUILD_RECORD.md` §25.5).

9.2 Code and data

Data and code live under `datasets/disaster_streams{,_hard,_sonnet5,_model}/`: `generate.py` (parametric streams, with a `--regime` switch), `realize.py` (LLM text via a batch API, with `--batch-id` recovery), `model_arm.py` (extraction, caching raw output for `--from-raw`), `extract.py` (surface parser and normalizer), and `evaluate.py` (tracker, baselines and gate). Evaluation:

```
uv run python datasets/disaster_streams/evaluate.py --data <dir> --split <val|test> \
    [--min-conf 0.99] [--reject-sigma 4] [--learn-gate]
```

Always pass the trivial baseline alongside; `evaluate.py` reports `est_last_value` beside `est_ekf` for the reason given in §4.

Known-negative settings, recorded so they are not retried: a symmetric 3σ innovation gate is catastrophic on rising tolls (use the one-sided, dynamics-aware gate); confidence-as-soft measurement noise does **not** beat a hard confidence cut, because zero-shot extraction errors are gross false positives rather than graded noise; and record-mode precision requires the boundary architecture, since the span decoder ignores structure mode.

The build record — attachment points, CLI, training plan, decision log, generator specification, the blind-test protocol, and the full negative-sampling implementation notes — is in `EKF_MHT_BUILD_RECORD.md`.

10. References

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